

Question #1 of 43

Question ID: 1472247

A bank in Canada is quoting CAD/USD 1.4950 – 1.5005, and USD/EUR 0.9350 – 0.9400. What is bid/ask exchange rate for CAD/EUR?

- A) CAD/EUR 1.3978 – 1.4105.
- B) CAD/EUR 0.6254 – 0.6264.
- C) CAD/EUR 1.5904 – 1.6048.



Explanation

The CAD/EUR bid quote is $1.495 \times 0.935 = 1.3978$

The CAD/EUR ask quote is $1.5005 \times 0.940 = 1.4105$

(Module 5.1, LOS 5.b)

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Question ID: 1472275

Under the Mundell-Fleming model and the portfolio balance approach to exchange rate determination, a country following sustained expansionary fiscal policy would see its currency:

- A) appreciate in the short-run and depreciate in the long-run.
- B) appreciate in the short-run and appreciate in the long-run.
- C) depreciate in the short-run and depreciate in the long-run.



Explanation

Under Mundell-Fleming model, a country running expansionary fiscal policy (i.e., running fiscal deficits) would attract foreign capital due to high interest rates and will see its currency appreciate in the short-run. Under the asset market approach, in the long-run sustained deficits will increase the risk of the country's debt and lead to a currency depreciation.

(Module 5.3, LOS 5.k)

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Question ID: 1508650

Today, the spot rate on pounds sterling is \$0.6960 and 90-day forward pounds are priced at \$0.6925. The forward discount/premium is:

- A) premium of \$0.0035.
- B) premium of \$0.0005.
- C) discount of \$0.0035.



Explanation

Premium (discount) = $F - S = 0.6925 - 0.696 = -0.0035$ (i.e., a discount).

(Module 5.1, LOS 5.a)

Question #4 of 43

Question ID: 1472255

Which of the following statements regarding relative purchasing power parity is *most* accurate? Relative purchasing power states that exchange rates:

- A) will change to reflect differences in inflation between countries.
- B) will change to reflect differences in nominal interest rates between countries.
- C) will change to reflect differences in real interest rates between countries.



Explanation

Purchasing power parity states that exchange rates will change to reflect differences in inflation between countries. Interest rate parity states that exchange rates must change so that risk-adjusted returns on investments in any currency will be equal.

(Module 5.2, LOS 5.e)

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Question ID: 1472279

Zimbaya is a developed economy with high capital mobility. Deborah Isaccson is evaluating the Zim (Z\$), the national currency of Zimbaya. Which of the following is *most likely* to lead to appreciation of Z\$? If Zimbaya starts following:

- A) an expansionary fiscal policy.
- B) a loose monetary policy.
- C) a restrictive fiscal policy.



Explanation

If Zimbaya follows an expansionary fiscal policy, government borrowing will increase leading to an increase in interest rates. This increase in interest rates will attract capital inflows to Zimbaya, leading to an appreciation of the Z\$. Either an expansionary ("loose") monetary policy or a restrictive fiscal policy should lead to lower interest rates and to depreciation of Z\$.

(Module 5.3, LOS 5.k)

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Question ID: 1472256

Which of the following statements regarding purchasing power parity (PPP) is *least* accurate?

- A) Under absolute PPP the foreign price level expressed in domestic currency terms should be equal to the domestic country's price level. 
- B) Absolute PPP is similar to the law of one price, except it concerns a basket of goods rather than a single good. 
- C) Relative PPP states that prices for goods and services are the same whether it is for one good or for a basket of goods. 

Explanation

Relative PPP does not state that prices for goods and services are the same, only that the rate of change in the FX rate is a function of the inflation differentials between the two countries.

(Module 5.2, LOS 5.e)

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Question ID: 1508652

Ninety days ago, Marc Samuelson entered into a 180-day forward contract to purchase 1 million CHF at an all-in rate of \$1.0225/CHF.

The following USD/CHF quotes are currently available in the market:

Spot	1.0301/1.0302
30 days	-12.3/-11.9
90 days	-16.0/-14.8
180 days	-18.9/-17.8

Interest rates:

90-day CHF	1.02%
180-day CHF	1.03%
90-day USD	1.00%
180-day USD	0.99%

The mark-to-market value of Samuelson's position is *closest* to:

- A) \$7,585.
- B) \$5,985.
- C) -\$6,735.



Explanation

Samuelson is currently long CHF in the forward market. Closing or offsetting that position requires a *short* forward contract in CHF (i.e., A contract to convert CHF into USD). To calculate the mark-to-market value, we first need to have the forward all-in bid price: $1.0301 - (16.0/10,000) = 1.0285$.

Mark-to-market value =

$$\frac{(FP_t - FP)(\text{contract size})}{(1 + R \frac{\text{days}}{360})} = \frac{(1.0285 - 1.0225)(1,000,000)}{(1 + (0.01(\frac{90}{360})))} = \$5,985$$

(Module 5.2, LOS 5.d)

Assume an investor living in Mauritius can borrow in \$ or in Mauritius Rupee (MUR). Given the following information, determine whether an arbitrage opportunity exists. If so, how much would the arbitrageur profit by borrowing MUR 1,000,000 or the equivalent in \$? (Assume a period of one year and state the profit in domestic currency terms.)

Spot rate (MUR/\$)	30.73000
Forward rate (MUR/\$)	31.50000
Domestic (MUR) interest rate (%)	6.50%
Foreign (\$) interest rate (%)	5.20%

Which of the following is *closest* to the correct answer?

A) Borrow domestic. Arbitrage profits are \$39,685.



B) Borrow MUR. Arbitrage profits are MUR 13,340.



C) Borrow \$. Arbitrage profits are \$13,340.



Explanation

Step 1: Determine whether an arbitrage opportunity exists.

- Using CIRP, $F = S \times (1+r_{\text{MUR}})/(1+r_{\$}) = 30.73 (1.065)/(1.052) = 31.11$ which is less than the market forward price of 31.50. Hence, \$ is overpriced in forward market.
- We sell \$ forward and purchase \$ in spot market

Step 2: Borrow/Lend which currency?

- Rule: Lend the currency that you are buying in the spot market and borrow the countercurrency

Step 3: Conduct Arbitrage and Calculate Profits.

Step	Description	Rate	Calculation	Result
a	Borrow MUR		MUR 1,000,000	MUR 1,000,000
b	Exchange MUR for \$	Spot	= MUR 1,000,000 / 30.73000 MUR/\$	\$32,541
c	Lend \$ at (U.S.) Rate		= \$32,541 × (1.05200)	\$34,233
d	Contract to sell proceeds fwd ¹	Fwd	= \$34,233 × 31.50000 MUR/\$	MUR 1,078,340
e	Calculate loan payoff ²		= MUR 1,000,000 × (1.06500)	MUR 1,065,000
f	Calculate profit (d-e)			MUR 13,340

Note: ¹ This is the amount you will have available to repay the loan. ² This is the amount you need to repay.

(Module 5.2, LOS 5.e)

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Question ID: 1472244

Given spot exchange rate of CAD/EUR 1.425-1.435, The spread is *closest* to:

A) 10 pips EUR



B) CAD 0.010



C) CAD 0.0010



Explanation

Spread = CAD 1.4350 – 1.4250 = CAD 0.010

(Module 5.1, LOS 5.a)

Jennifer Nance has recently been hired as an analyst at the Central City Bank in the currency trading department. Nance, who recently graduated with a degree in economics, will be working with other analysts to determine if there are profit opportunities in the foreign exchange market.

Nance has the following data available:

	U.S. Dollar (\$)	U.K. Pound (£)	Euro(€)
Expected inflation rate	6.0%	3.0%	7.0%
One-year nominal interest rate	10.0%	6.0%	9.0%

Market Spot Rates			
	U.S. Dollar (\$)	U.K. Pound (£)	Euro(€)
U.S. Dollar (\$)	\$1.0000	\$1.6000	\$0.8000
U.K. Pound (£)	0.6250	1.0000	2.0000
Euro (€)	1.2500	0.5000	1.0000

Market 1-year Forward Rates			
	U.S. Dollar (\$)	U.K. Pound (£)	Euro(€)
U.S. Dollar (\$)	\$1.0000	\$1.6400	\$0.8082
U.K. Pound (£)	0.6098	1.0000	2.0292

Euro (€)	1.2373	0.4928	1.0000
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Nance receives a report from Jamshed Banaji, Chief Economist at Central City Bank providing broad U.K. and U.S. macro-economic forecasts. Nance notes that the Bank of England is expected to pursue an expansionary monetary policy while the Federal Reserve monetary policy is expected to be neutral. Also, the British parliament is expected to reduce the budget deficits more aggressively as compared to the U.S.

Question #10 - 13 of 43

Question ID: 1472262

The no-arbitrage one-year forward USD/EUR rate is *closest* to:

- A) USD/EUR 0.7925.
- B) USD/EUR 0.8082.
- C) USD/EUR 0.8073.



Explanation

Interest rate parity implies that, in order to prevent covered interest arbitrage, the one-year forward USD/EUR rate should be equal to $\$0.8000(1.10) / (1.09) = \0.8073 .

(Module 5.2, LOS 5.e)

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Question ID: 1472263

For this question only, assume that the United States has a current account surplus versus the U.K. The amount by which the £/\$ has to change to restore current account balance is *least likely* to depend on:

- A) the initial level of current account surplus.
- B) the projected current account deficit.
- C) the response of import and export demand to changes in export prices.



Explanation

The adjustment to exchange rates to restore current accounts to a balanced position depends on:

- The level of initial deficit.
- The response of import and export demand to changes in import and export prices.
- The response of import and export prices to changes in the exchange rate.

(Module 5.2, LOS 5.e)

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Question ID: 1472264

For an investor pursuing a carry-trade, the funding currency would *most likely* be the:

- A) Pound. 
- B) Euro. 
- C) U.S. Dollar. 

Explanation

Under a carry trade, the funding currency is the lower yielding currency (in this case, the pound with 1-year nominal interest rate of 6% is the best candidate).

(Module 5.2, LOS 5.e)

Question #13 - 13 of 43

Question ID: 1472265

Which of the following is *least likely* to be a warning sign for currency crisis?

- A) Real exchange rate substantially lower than mean reverting level. 
- B) Nominal credit relative to bank reserves increase. 
- C) Inflation increases. 

Explanation

One of the warning signs of a currency crisis is that real exchange rate is substantially higher than the mean reverting level.

(Module 5.2, LOS 5.e)

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Question ID: 1472258

If the one-year forward exchange rate is DC/FC 2 and the spot rate is DC/FC 1.9 when the foreign rate of return is 12% and the domestic return is 10%, which of the following statements would be *most* accurate?

A) Arbitrage is possible here, investors should borrow domestic, lend foreign. 

B) Arbitrage is possible here, investors should borrow foreign, lend domestic. 

C) The arbitrage possibilities cannot be determined with the data given. 

Explanation

Question 1: Is there an arbitrage opportunity?

If the result of the following formula (derived from rearranging the interest rate parity condition) is *not* equal to 0, there is an arbitrage opportunity.

$$(1 + r_{\text{domestic}}) - [(1 + r_{\text{foreign}}) \times \text{Forward}_{\text{DC/FC}}] / \text{Spot}_{\text{DC/FC}} = ?$$

Here, $(1 + 0.10) - [(1 + 0.12) \times 2.0_{\text{DC/FC}}] / 1.9_{\text{DC/FC}} = (1.10 - 1.18) = -0.08$, which is not equal to 0. **Arbitrage opportunities exist.**

Question 2: Borrow Domestic (local) or Foreign?

Here are some "rules" regarding where to start the arbitrage (where to borrow). *These rules only work if there are no transaction costs and only if the currency is quoted in DC/FC terms.*

Rule 1: If the sign on the result of question 1 is *negative*, *borrow domestic*. If the sign is positive, borrow foreign. Here, the sign is negative, so borrow domestic.

Rule 2:

$(r_d - r_f) < (\text{Forward} - \text{Spot}) / \text{Spot}$ then Borrow Domestic

$(r_d - r_f) > (\text{Forward} - \text{Spot}) / \text{Spot}$ then Borrow Foreign

Here, borrow domestic:

$$(r_d - r_f) = (0.10 - 0.12) = -0.02 < (\text{Forward} - \text{Spot}) / \text{Spot} = (2.0_{\text{DC/FC}} - 1.9_{\text{DC/FC}}) / 1.9_{\text{DC/FC}} = 0.05$$

$$-0.02 < 0.05$$

Summary: To take advantage of arbitrage opportunities, borrow domestic and lend foreign.

(Module 5.2, LOS 5.e)

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Question ID: 1472259

One-year interest rates are 7.5% in the U.S. and 6.0% in New Zealand. The current spot exchange rate is USD/NZD 0.5500. If uncovered interest rate parity holds, the expected spot rate in one year must be *closest* to:

A) USD/NZD 0.54233.



B) USD/NZD 0.55825.



C) USD/NZD 0.56675.



Explanation

USD interest rate is 1.5% higher hence, NZD will appreciate by 1.5% under the uncovered interest rate parity.

Expected Spot = $0.5500 \times (1.015) = \text{USD/NZD } 0.55825$

(Module 5.2, LOS 5.e)

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Question ID: 1508654

Which of the following is *least likely* the objective of central bank intervention?

A) have ability to pursue an independent monetary policy.



B) prevent appreciation of domestic currency.



C) reduce excessive inflow of foreign capital.



Explanation

Central bank objectives include prevention of *excessive* appreciation of domestic currency, reduction of excessive foreign capital inflows and pursuit of independent monetary policy.

(Module 5.3, LOS 5.I)

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Question ID: 1472273

Tim Kramer is assessing the risks of the carry trade for his firm. He obtains a distribution of expected returns for the carry trade. This distribution is *most likely* to exhibit:

A) fat tails and a negative skew.



B) a normal distribution.



C) fat tails and a positive skew.



Explanation

The distribution of carry trade returns is characterized by *negative* skewness and *fat tails*.
(Module 5.3, LOS 5.i)

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Question ID: 1472271

The return distribution of FX carry trade is characterized by:

A) negative skewness and negative excess kurtosis.



B) negative skewness and positive excess kurtosis.



C) positive skewness and positive excess kurtosis.



Explanation

FX carry trade return distribution exhibits negative skewness and positive excess kurtosis.
(Module 5.3, LOS 5.i)

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Question ID: 1472269

Assume that the domestic nominal rate of return is 4% and the foreign nominal rate of return is 5%. If the current exchange rate is DC/FC 0.400, the forward rate consistent with covered interest rate parity is:

A) 0.400.



B) 0.318.



C) 0.396.



Explanation

$F/S = (1 + r_D) / (1 + r_F)$ where the currency is quoted as DC/FC

$$F = (1.04/1.05)(0.400) = 0.396$$

(Module 5.2, LOS 5.g)

The following information is gathered for three countries:

Country	Comment
A	Current account deficit is very large relative to GDP
B	Imports highly price-elastic goods
C	Exports global commodities

Which country will *most likely* see its current account deficit restored to sustainable level more rapidly under the flows mechanism of balance of payments?

A) Country A.



B) Country B.



C) Country C.



Explanation

Countries with lower initial current account deficits, with import and export prices sensitive to exchange rate movements and with imports and exports with high price elasticity of demand would see their current account deficits quickly restored to sustainable level due to depreciation of their currency. Country B imports goods that have high price elasticity. Country A has large current account deficit and hence will take time to adjust to sustainable level. Country C exports commodities whose global prices are not sensitive to their own currency's values.

(Module 5.3, LOS 5.j)

Terrance Burnhart, a junior analyst at Wertheim Investments Inc., was discussing the concepts of purchasing power parity (PPP) and uncovered interest rate parity (UIRP) with his colleague, Francis Ferngood. During the conversation Burnhart made the following statements:

Statement 1: Absolute PPP is based on a number of unrealistic assumptions that limits its real-world usefulness. These assumptions are: that all goods and services can be transported among countries at no cost; and all countries use the same basket of goods and services to measure their price levels.

Statement 2: UIRP rests on the idea of equal real interest rates across international borders. Real interest rate differentials would result in capital flows to the higher real interest rate country, equalizing the rates over time. Another way to say this is that differences in interest rates are equal to differences in expected changes in exchange rates.

With respect to these statements:

A) only statement 1 is correct.



B) both are correct.



C) only statement 2 is correct.



Explanation

UIRP means that interest rates and exchange rates will adjust so the risk adjusted return on assets between any two countries and their associated currencies will be the same. PPP is based on the idea that a given basket of goods should cost the same in different countries after taking into account the changes in exchange rates. PPP does not hold due to transportation costs and other factors.

(Module 5.2, LOS 5.f)

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Question ID: 1472260

The domestic interest rate is 9% and the foreign interest rate is 7%. If the forward exchange rate is DC/FC 5.00, what spot exchange rate is consistent with covered interest parity?

A) 4.91.



B) 5.09.



C) 4.83.



Explanation

$$\text{Forward}_{\text{DC/FC}} / \text{Spot}_{\text{DC/FC}} = (1 + r_{\text{domestic}}) / (1 + r_{\text{foreign}}).$$

$$\text{Spot}_{\text{DC/FC}} = \text{Forward}_{\text{DC/FC}} (1 + r_{\text{foreign}}) / (1 + r_{\text{domestic}}) = (5.00)(1.07) / (1.09) = 4.908$$

(Module 5.2, LOS 5.e)

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Question ID: 1472272

The carry trade is *most likely* to be profitable when:

- A) uncovered interest rate parity holds. 
- B) the forward rate is biased estimator of future spot rate. 
- C) the Fisher relation is violated. 

Explanation

The carry trade is premised on uncovered interest rate parity *not* holding. When the forward rate is an *unbiased* predictor of the future spot rate, uncovered interest rate parity will hold and hence the carry trade will not be profitable. When the forward rate is a *biased* predictor of future spot rate, uncovered interest rate parity will not hold and the carry trade may be profitable.

(Module 5.3, LOS 5.i)

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Question ID: 1472270

Given the following information, what is the forward exchange rate implied by interest rate parity?

- U.S. interest rate = 9%.
- North Korea interest rate = 10%.
- Spot rate = 1.65 KPW/\$.

- A) 1.665 KPW/\$. 
- B) 1.635 KPW/\$. 
- C) 0.612 KPW/\$. 

Explanation

Forward rate (DC/FC) = Spot Rate (DC/FC) $\times [(1 + \text{domestic rate}) / (1 + \text{foreign rate})]$,

Forward rate = $1 / 1.65 \text{ (KPW/\$)} \times (1.09 / 1.10) = 0.60055 \text{ \$/KPW}$, or $1.665 \text{ KPW/\$}$.

Alternatively, forward rate = $1.65 \text{ (KPW/\$)} \times (1.10 / 1.09) = 1.665 \text{ (KPW/\$)}$.

(Module 5.2, LOS 5.g)

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Question ID: 1472278

Under high capital mobility, the Mundell-Fleming model to determine exchange rate focuses on the impact of:

A) trade balance.



B) interest rates.



C) inflation.



Explanation

Mundell-Fleming approach focuses on the role of interest rate in exchange rate determination. Mundell-Fleming model does not explicitly take into account the role of inflation.

(Module 5.3, LOS 5.k)

Patrick Sheehan is the head of foreign currency desk at GPN Bank NA, a large U.S. Bank holding company. Patrick is concerned about recent spike in volatility of EUR. He obtains current spot and forward quotes from his terminal (given in Exhibit 1). He also collects interest rate information (given in Exhibit 2).

Exhibit 1: Current spot and forward exchange rate quotes

Currency Paid	Spot rates	Forward rates		
		30-day	60-day	90-day
USD/EUR	1.3110–14	+3.18/+3.38	+6.73/+7.18	+10.48/+10.78
CHF/USD	0.9273–77	–4.09/–3.79	–8.45/–7.95	–12.80/–12.05
USD/GBP	1.6242–47	–26.10/–24.6	–50.20/–47.20	–72.20/–68.2

Exhibit 2: Selected interest rates

Interest Rates	USD	EUR	CHF	INR
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30 day	0.21%	0.90%	1.12%	6.72%
60 day	0.22%	0.93%	1.15%	6.84%
90 day	0.25%	1.04%	1.25%	6.90%

Sheehan reviews bank's current open forward contracts. One of the contracts calls for purchase of EUR 200 million at an all-in rate of USD 1.3912 and matures in 30 days.

During the market turmoil of late 2008, GPN had lost a lot of money in FX carry trades. Sheehan realizes that GPN has not established any new FX carry trade positions since then and is anxious to establish new positions. One trade that he finds promising is a carry trade in Indian Rupee (INR). Sheehan notes that while the spot rate is 53.88 INR/USD, Melissa Andrews, GPN's Chief Economist expects the Rupee to trade at 54.12 INR/USD in 90 days.

During his conversation with Andrews, Sheehan asks about the driving factor behind depreciation of the Rupee in the recent past. Andrews explains that there are several theories to explain exchange rate movements. Personally, she prefers to focus on the long-term implications of fiscal policy. She feels that the interest rate in India is expected to be higher than in the U.S. She also states that India is following a more expansionary fiscal policy as compared to the U.S. and that policy is expected to continue. Sheehan observes that such deficits have resulted in large external debt relative to GDP for India.

Sheehan observes that over the past decade, capital controls in India have been loosened resulting in free flow of capital. Additionally, due to a relatively more restrictive monetary policy in India relative to the U.S., nominal interest rates have been substantially higher in India as compared to the U.S.

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Question ID: 1586016

The current mark-to-market value of the EUR forward contract is *closest* to:

- A) -USD 15,889,620.
- B) USD 15,976,000.
- C) -USD 15,973,205.



Explanation

The contract calls for purchase of 200 million EUR in 30 days. To compute the mark-to-market value, we would have to use the quote on 30-day forward contract to sell EUR. Given USD/EUR quote structure, we should use the bid price (going up the quote).

$$V_t = \frac{(FP_t - FP)(\text{Contract size})}{\left[1 + R\left(\frac{\text{Days}}{360}\right)\right]}$$

$$FP_t = 1.3110 + 3.18/10,000 = 1.31132$$

$$FP = 1.3912 \text{ (given)}$$

R = 30-day USD interest rate (we are discounting USD, hence use U.S. interest rate) = 0.21%

$$V_t = \frac{(1.31132 - 1.3912)(200,000,000)}{\left[1 + 0.0021\left(\frac{30}{360}\right)\right]} = \frac{-15,976,000}{1.000175} = -15,973,205$$

(Module 5.3, LOS 5.k)

Question #27 - 29 of 43

Question ID: 1586017

Regarding the valuation of INR, Andrews would *most likely* use:

A) Monetary approach.



B) Portfolio Balance Approach.



C) Mundell-Fleming model.



Explanation

Portfolio balance approach focuses on long-term implications of fiscal policy on exchange rate. Monetary approach focuses on implications of monetary policy while Mundell-Fleming model focuses on short-term implications of monetary/fiscal policies.

(Module 5.3, LOS 5.k)

Question #28 - 29 of 43

Question ID: 1586018

Based on the Mundell-Fleming model, relative to the USD, the INR would *most likely*:

A) depreciate.



B) appreciate.



C) appreciate or depreciate.



Explanation

Under the Mundell-Fleming model, given high capital mobility, an expansionary fiscal policy combined with a restrictive monetary policy would lead to appreciation of the INR in the short-term.

(Module 5.3, LOS 5.k)

Question #29 - 29 of 43

Question ID: 1586019

Under the portfolio balance approach to exchange rate determination, relative to USD, INR would *most likely*:

- A) appreciate in the short-term.
- B) appreciate in the long-term.
- C) depreciate in the long-term.



Explanation

Under the portfolio balance approach, large levels of debt would lead to currency depreciation in the long-term.

(Module 5.3, LOS 5.k)

Question #30 of 43

Question ID: 1472249

Donna Ackerman, CFA, is an analyst in the currency trading department at State Bank. Ackerman is training a new hire, Fred Bos, a recent college graduate with a BA in economics.

Ackerman and Bos have the following information available to them:

Spot Rates		
	Bid Price	Ask Price
EUR/USD	€1.0000	€1.0015
GBP/USD	£2.0000	£2.0100
EUR/GBP	€0.3985	€0.4000

Ackerman and Bos are interested in pursuing profitable arbitrage opportunities for State Bank. What will be the profits from triangular arbitrage, starting with \$1,000?

A) \$243.78.



B) \$245.65.



C) \$248.46.



Explanation

The EUR/USD and GBP/USD rates imply that the arbitrage free cross rates for the EUR/GBP are:

$$\text{bid} = \text{€}1.000 / \text{£}2.0100 = \text{€}0.4975$$

$$\text{ask} = \text{€}1.0015 / \text{£}2.0000 = \text{€}0.5008$$

Since the cross rates given ($\text{€}0.3985 - \text{€}0.4000$) are outside of the arbitrage-free cross rates, profitable arbitrage is available.

It takes too few euros to buy 1 pound, so we want our arbitrage trades to go in the direction that will cause us to sell overvalued euros for pounds at the ask rate of $\text{€}0.4000$.

Start with \$1,000.

Use the \$1,000 to buy euros ($\$1,000 \times \text{€}1.000/\$$) = $\text{€}1,000$.

Use the $\text{€}1,000$ to buy sterling ($\text{€}1,000 / \text{€}0.4000/\text{£}$) = $\text{£}2,500$.

Use the $\text{£}2,500$ to buy dollars ($\text{£}2,500 / \text{£}2.0100/\$$) = $\$1,243.78$.

(Module 5.1, LOS 5.b)

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Question ID: 1472276

Country P has high capital mobility and has recently switched from a balanced fiscal policy to an expansionary fiscal policy. Over time this expansionary fiscal policy is expected to lead to an increase in the government debt-to-GDP ratio.

If we simultaneously consider both the Mundell-Fleming and the portfolio balance model, in the short term country P's currency is *most likely* to:

A) depreciate.



B) remain stable.



C) appreciate.



Explanation

Under the *Mundell-Fleming* model, country P's currency should appreciate in the short-term as fiscal deficits push interest rates higher. Under the *portfolio balance* model, such a government that runs a large budget deficit should *over time* see a decline in the country's currency value – however, this is a *long-run* rather than *short-run* effect.

(Module 5.3, LOS 5.k)

Question #32 of 43

Question ID: 1508651

An investor has entered into a 90-day forward contract to purchase 2 million GBP at an all-in rate of USD 1.4612. In 30 days, the following quotes were available:

	USD/GBP
spot rate	1.4522–24
30-day forward rate	1.4618–21
60-day forward rate	1.4621–25
90-day forward rate	1.4632–36

Interest rate information:

Interest rates	When contract was initiated		Currently (t=30)	
	USD	GBP	USD	GBP
30-day	0.20%	0.32%	0.20%	0.32%
60-day	0.21%	0.32%	0.21%	0.32%
90-day	0.21%	0.33%	0.21%	0.33%

The mark-to-market value of the forward contract is *closest* to:

A) USD 1999.



B) USD 1800.



C) USD 2599.



Explanation

To unwind the forward contract, the investor would enter into a 60-day forward contract to *sell* GBP. The relevant exchange rate is 1.4621. The value obtained will be in price currency (USD) and would be discounted at USD interest rate for 60 days (at t=30).

$$V_t = \frac{(FP_t - FP)(\text{contract size})}{\left[1 + R\left(\frac{\text{days}}{360}\right)\right]} = \frac{(1.4621 - 1.4612)(2,000,000)}{\left[1 + 0.0021\left(\frac{60}{360}\right)\right]} = 1799$$

(Module 5.2, LOS 5.d)

Question #33 of 43

Question ID: 1472283

Which of the following is *least likely* a warning sign of an impending currency crisis?

- A) Liberalized capital markets that allow for a free flow of capital.
- B) Terms of trade deteriorate.
- C) Money supply relative to bank reserves shrinks.



Explanation

Warning sign of an impending currency crisis is when money supply relative to bank reserves grows (not shrinks).

(Module 5.3, LOS 5.m)

Question #34 of 43

Question ID: 1472246

Which of the following statements about foreign currency bid-ask spreads is *least* accurate?
Foreign currency bid-ask spreads:

- A) are influenced by time window in a trading day.
- B) increase as the size of the transaction decreases.
- C) are a function of transaction volume and volatility.



Explanation

Bid-ask spreads are size related in that the larger the transaction the *larger* the spread.

(Module 5.1, LOS 5.a)

Question #35 of 43

Question ID: 1472257

Given currency quotes in DC/FC, if: $1 + r_{DC} < (1 + r_{FC}) \times (\text{forward rate} / \text{spot rate})$ funds will:

- A) flow out of the domestic country.
- B) flow into the domestic country.
- C) flow in and out of the domestic country.



Explanation

This equation is Interest Rate Parity rearranged! If the term on the left ($1 + r_{DC}$), is less than the term on the right, it means that the domestic rate is low relative to the hedged foreign rate. Therefore, there is a profitable arbitrage from borrowing the domestic currency and lending at the foreign interest rate. Because we lend in the foreign market, we say that the funds flow out of the domestic economy.

(Module 5.2, LOS 5.e)

Question #36 of 43

Question ID: 1472248

Given the following quotes for the Canadian dollar (CAD) and the British pound (GBP), the implied CAD/GBP bid-ask quotes are *closest* to:

CAD/USD 1.59031 – 1.59701

GBP/USD 0.69459 – 0.69686

A) CAD/GBP 2.2895 – 3.2886.



B) CAD/GBP 2.2821 – 2.2992.



C) CAD/GBP 2.2992 – 2.3163.



Explanation

USD/GBP(bid) = $1/0.69686 = 1.4350$

USD/GBP(ask) = $1/0.69459 = 1.4397$

CAD/GBP bid quote is $1.4350 \times 1.5903 = 2.2821$

CAD/GBP ask quote is $1.4397 \times 1.5970 = 2.2992$

(Module 5.1, LOS 5.b)

Question #37 of 43

Question ID: 1508655

Which of the following is *least likely* a warning sign of an impending currency crisis?

A) Official foreign exchange reserves decline dramatically.



B) Currency value is substantially higher than the mean-reverting level.



C) Floating exchange rates.



Explanation

Warning sign of an impending currency crisis is when exchange rates are fixed or partially fixed (and not floating).

(Module 5.3, LOS 5.m)

Question #38 of 43

Question ID: 1472266

Ackerman explains to Bos that a theoretical relationship exists between forward rates and future spot rates, called the forward rate parity. This relation suggests that:

- A) the forward rate is an unbiased predictor of the expected future spot rate, and uncovered interest rate parity would hold. 
- B) the forward rate is a biased predictor of the expected future spot rate, and uncovered interest rate parity would not hold. 
- C) the forward rate is an unbiased predictor of the expected future spot rate, and uncovered interest rate parity would not hold. 

Explanation

The forward rate parity is $F = E(S_1)$, meaning that the forward rate is an unbiased predictor of the expected future spot rate. If this is the case, uncovered interest rate parity would be same as covered interest rate parity and since covered interest rate parity holds (by arbitrage), uncovered interest rate parity would also hold.

(Module 5.2, LOS 5.f)

Question #39 of 43

Question ID: 1472250

The forward rate on a 90-day contract is FC/USD 5 and the spot is FC/USD 4. The USD is trading at a forward:

- A) premium of 0.8. 
- B) discount of 1.0. 
- C) premium of 1.0. 

Explanation

Base currency (USD in this case) is at a forward premium if the forward rate is above the spot rate. Forward premium = forward rate – spot rate = $5 - 4 = 1$.

(Module 5.1, LOS 5.c)

Question #40 of 43

Question ID: 1472267

Professor Imada Suzaken made the following statement to his economics class: "If you can earn 8% on A-rated bonds in the U.S. but only 6% on similar bonds in Canada, Canadian investors may want to buy those bonds in the U.S. for the excess return. However, after collecting the extra dollars, the investors would lose those profits when they converted their gains into their home currency." Suzaken's statement *most accurately* describes:

- A) covered interest rate parity. 
- B) uncovered interest-rate parity. 
- C) purchasing-power parity. 

Explanation

Uncovered interest-rate parity is the concept that exchange rates must change so that the return on investments with identical risk will be the same in any currency. Suzaken's statement reflects uncovered interest rate parity. Covered interest rate parity would be applicable if the investor hedges the foreign exchange risk via a forward exchange rate contract.

(Module 5.2, LOS 5.f)

Question #41 of 43

Question ID: 1472280

Ashok Jain is assessing the currency value of Lutina. Jain believes that prices are sticky in the short term and, hence, do not immediately reflect changes in monetary policy. If Lutina announces a change to a restrictive monetary policy, Jain would *most likely* conclude that Lutina's currency would:

- A) excessively depreciate in the long-term. 
- B) excessively appreciate in the long-term. 
- C) excessively appreciate in the short-term. 

Explanation

Dornbusch overshooting model. This model assumes that prices are sticky (inflexible) in the short term and, hence, do not immediately reflect changes in monetary policy.

The model concludes that exchange rates will overshoot the long-run PPP value in the short term. A restrictive monetary policy leads to excessive appreciation of the domestic currency in the short term and then a slow depreciation toward the long-term PPP value.

(Module 5.3, LOS 5.k)

Question #42 of 43

Question ID: 1472253

Assume an investor living in Japan can borrow in the domestic yen (JPY) or in the foreign U.S. dollar (USD). Given the following information, determine whether an arbitrage opportunity exists. If so, how much would the investor profit by borrowing JPY 58,175,000 or the equivalent in USD? (Assume a period of one year.)

Spot rate (JPY/USD)	116.35
Forward rate (JPY/USD)	112.99
Domestic (Japanese) interest rate (%)	1.50
Foreign (U.S.) interest rate (%)	4.00

- A)** An arbitrage opportunity results in a profit of JPY 292,825.
- B)** An arbitrage opportunity results in a profit of JPY 27,963.
- C)** An arbitrage opportunity results in a profit of JPY 25,170.

**Explanation**

Step 1: Determine whether an arbitrage opportunity exists.

We can arrange the formula for covered interest rate parity to look like:

$$(1 + r_{\text{domestic}}) - [((1 + r_{\text{foreign}}) \times \text{Forward}_{\text{DC/FC}}) / \text{Spot}_{\text{DC/FC}}] = 0$$

If this condition holds with the financial data above, there are no arbitrage opportunities.

$$(1 + 0.01500) - [((1 + 0.04000) \times 112.99000) / 116.35000] = 1.01500 - 1.00997 = \mathbf{0.00503}$$

Since the no arbitrage condition does not hold, we move on to:

Step 2: Borrow Domestic or Foreign?

The sign on the result of step 1 is positive, so borrow foreign.

$(r_d - r_f)$		$(\text{Forward} - \text{Spot}) / \text{Spot}$
$(0.01500 - 0.04000)$		$(112.99000 - 116.35000) / 116.35000$
-0.02500	>	-0.02888

Step 3: Arbitrage Process

Description	Rate	Calculation	Result
Calculate foreign equivalent & borrow this amount.	Spot	JPY 58,175,000 / 116.35000JPY/USD	USD 500,000
Invest Domestic at Domestic interest rate*		JPY 58,175,000 × (1 + 0.01500)	JPY 59,047,625
* This is the amount you will have available to repay the loan.			
Calculate loan payoff (foreign currency)		500,000USD × (1 + 0.04000)	USD (520,000)
Calculate payoff in Domestic currency**	Fwd	520,000USD × 112.99000JPY/USD	JPY (58,754,800)
**This is the amount you need to repay.			
Calculate Arbitrage Profit		JPY 59,047,625 – JPY 58,754,800	JPY 292,825

(Module 5.2, LOS 5.e)

Country P has high capital mobility and has recently switched from balanced fiscal policy to an expansionary fiscal policy. Over time this expansionary is expected to lead to an increase in government debt to GDP ratio.

If we simultaneously consider both the Mundell-Fleming and the portfolio balance model, in the long run country P's currency is *most likely* to:

A) appreciate.



B) depreciate.



C) remain stable.



Explanation

Under the portfolio balance model, as the ratio of government debt to GDP increases over time and the level of government debt becomes unsustainable, the currency of country P should depreciate. (Under the Mundell-Fleming model, country P's currency should appreciate *in the short-term* as fiscal deficits push interest rates higher, however this question is specifically asking about the long-run effect).

(Module 5.3, LOS 5.k)